

Tempus What's New

Product Version 26.10

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About This Manual

This manual provides information about Cadence® Tempus™ Timing Solution.

How This Document Is Organized

This manual is organized into chapters that cover broad areas of Tempus functionality. Each chapter contains topics that address one or more of the following areas:

- New functionality in Tempus and enhancements made to existing forms and commands to support a new feature.
- Changes in default behavior, name changes to existing commands and forms, and syntax changes.
- Major documentation changes, such as a new chapter or substantial reorganization.

Related Documents

For more information about Tempus, see the following documents. You can access these and other Cadence documents with the Cadence Help online documentation system.

- [*Tempus Known Problems and Solutions*](#)
Describes important Cadence Change Requests (CCRs), and provides workarounds for known problems.
- [*Tempus Text Command Reference*](#)
Describes the Tempus text commands, including syntax and examples.
- [*Tempus Menu Reference*](#)
Describes the Tempus graphical user interface.
- [*Tempus User Guide*](#)
Provides information about using various Tempus features.
- [*Tempus Foundation Flows User Guide*](#)
Describes the Cadence-recommended procedures for performing basic STA and SI signoff, MMMC signoff, and CPF-based MMMC signoff using Cadence® Tempus™ Timing Solution.
- README file
Contains installation, compatibility, and other prerequisite information, including a list of Cadence change requests (CCRs) that were resolved in this release. You can read this file online at downloads.cadence.com.

Release Overview

- Release 26.10 Overview
 - Release Highlights
 - Timing Analysis Enhancements
 - GBA Pessimism Reduction Improvements
 - Hierarchical Flow Enhancements
 - Runtime and Memory Improvements
 - Accuracy Improvements
 - Infrastructure Enhancements
 - New Text Commands and Global Variables
 - New Command Parameters
 - Obsolete Commands
 - Removed from Software
 - Obsolete Global Variables
 - Removed from Software
 - Obsolete Command Parameters
 - Removed from Software

Release 26.10 Overview

Release Highlights

Timing Analysis Enhancements

Derate-Based Advanced Multi-Input Switching (AMIS-Derate) Analysis

The new analysis feature, Derate-based AMIS (AMIS-Derate), is now available in this release. A simulation-based advanced multiple-input switching (AMIS-Simulation) analysis, already available in Tempus previous releases, can model MIS effects on both delay and slew. As a result of modeling MIS effects on slew, enabling AMIS-simulation can be very pessimistic. Due to the detailed simulation performed for each instance, the runtime overhead can be high.

To mitigate these shortcomings, the AMIS-derate feature has been introduced. AMIS-derate pre-characterizes MIS impact on multi-input cells as delay derates for different slew/load combinations and uses these derates in timing analysis. AMIS-derate applies MIS impact only to the base delay component. AMIS-derate maintains accuracy similar to AMIS-simulation and provides better runtime, debuggability, and reporting capabilities.

For more information, refer to "[Support for Derate-Based Advanced Multi-Input Switching \(AMIS-Derate\) Analysis](#)".

Virtual IDX (VIDX) SPEF Support in Tempus 3DIC Analysis

Tempus STA now supports reading VIDX SPEF generated by Quantus. The VIDX SPEF models the capacitance impact between two dies.

GBA Pessimism Reduction Improvements

Arc Bounding Box Mode for Spatial Derates

Spatial derate calculation uses the `chip_size` mode in GBA, which is conservative. The new arc bounding box mode has been introduced in Tempus STA, which reduces GBA spatial derate pessimism by limiting bounding box calculations to all timing paths through an instance. The results in reduced GBA-PBA gap and reduced GBA pessimism result in improved PBA runtime.

Improved GBA Waveform Compression

The enhanced GBA waveform compression reduces pessimism in graph-based analysis by refining the transition waveform representation to better reflect true signal behavior. This preserves waveform accuracy while minimizing conservatism. And also results in a reduced GBA-PBA gap.

Hierarchical Flow Enhancements

Interface Timing Context Generation in DSTA

Tempus now supports timing context generation in DSTA, enabling interface path reporting in hierarchical context analysis. Previously in DSTA, it was limited only to register-to-register path reporting.

Pushdown SDC Support in DSTA

Tempus supports Pushdown SDC generation in DSTA and enables both register-to-register and interface path reporting in hierarchical context analysis.

Boundary Model Generation without Additional Timer Update

Tempus has been enhanced to support boundary models generation without triggering a timer update internally. Previously, boundary models generation implicitly triggered a timer update, resulting in performance overhead. This enhancement is primarily beneficial for hierarchical flows that require both timing context and boundary models generation in the same run.

Runtime and Memory Improvements

Faster `update_timing` with Selective in-Memory SI Reporting

This enhancement improves the `update_timing` command performance by optimizing single-CPU runtime and multi-CPU scalability with the `set_si_mode -enable_delay_report true` setting.

New Advanced SI Analysis

The new advanced SI analysis changes how attacker slews are computed by establishing a signoff accurate base delay calculation before SI analysis begins. This replaces the earlier approach that relied on a fast pre-SI analysis attacker slew computation. Beyond the accuracy benefits, it also offers runtime benefits from enhanced reuse provided by signoff-accurate base iteration.

Accuracy Improvements

Consolidated Delay Calculation Framework

This enhancement in delay calculation aligns key simulation steps within a single, consistent computation framework across timing and SI analysis. By synchronizing these steps, the analysis achieves improved accuracy, greater robustness, and better correlation with SPICE results.

Infrastructure Enhancements

Python Shell Support

In this release, Python Shell is supported in Tempus STA. This can be used for data analysis, visualization, and algorithm development. Python shell can be accessed in two ways:

- **Dedicated Python shell mode:** In this mode, the Tempus console is changed into a Python interpreter where Python code can be specified directly. All the Tempus Python commands are available as Python functions.
- **Embedded Python execution (within Tcl):** In this mode, blocks of Python code can be inserted within Tcl scripts using the `eval_python {}` command. This mode can be used to leverage Python's data processing and scripting capabilities for specific tasks in a Tcl-based flow.

Parallel report_timing (PRT) Framework

Tempus STA now supports a robust and flexible framework that allows parallel execution of multiple `report_timing` commands. This helps to reduce overall reporting runtime when the right set of `report_timing` commands is specified for PRT execution.

New Text Commands and Global Variables

The following table lists the commands and global variables that were added to the Tempus software. The second column identifies the chapter of the *Tempus Text Command Reference* where the command is documented.

New Commands and Global Variables	Chapter
<code>get_via_variation_mode</code>	Signal Integrity Commands
<code>merge_noise_reports</code>	Signal Integrity Commands
<code>restore_db_keep_updated_macro_lefs</code>	Import and Export Global Variables
<code>save_db_restrict_under_eco_batch_mode</code>	Import and Export Global Variables
<code>set_advanced_variation_mode</code>	Signal Integrity Commands
<code>set_via_variation_mode</code>	Signal Integrity Commands

<code>timing_consider_late_arrival_for_ssi_derate</code>	Global Configuration Variables
<code>timing_disable_get_clocks_for_current_instance</code>	Global Configuration Variables
<code>timing_enable_hierarchical_get_nets_support</code>	Global Configuration Variables
<code>timing_extract_model_enable_output_pin_attributes</code>	Global Configuration Variables
<code>timing_extract_model_half_cycle_non_interface_paths</code>	Global Configuration Variables
<code>timing_extract_model_synchronous_clocks_as_skew_checks</code>	Global Configuration Variables
<code>timing_extract_model_write_function_for_combinational_arcs</code>	Global Configuration Variables
<code>timing_ipd_xdomain_path_filtering_threshold</code>	Global Configuration Variables
<code>timing_library_ignore_combinational_arc_mismatch_tables</code>	Global Configuration Variables
<code>timing_property_is_clock_pin_consider_clock_source</code>	Global Configuration Variables
<code>timing_report_check_timing_enable_clock_crossing_verbose_clock_groups</code>	Global Configuration Variables
<code>timing_report_constraint_enable_drv_on_ideal_network</code>	Global Configuration Variables

<code>timing_report_enable_qpin_as_start_for_begin_end_pair</code>	Global Configuration Variables
<code>timing_report_enable_write_twf_third_party_for_input_pins</code>	Global Configuration Variables
<code>timing_report_limit_max_path_count</code>	Global Configuration Variables
<code>timing_report_print_normalized_slack</code>	Global Configuration Variables
<code>timing_report_skip_icg_from_all_fanin_fanout_collection</code>	Global Configuration Variables

New Command Parameters

The following table lists the parameters that were added to the Tempus Timing Solution software. The second column identifies the chapter of the Tempus Text Command Reference where the command is documented.

New Parameters	Chapter
<code>add_net</code> <code>-module_based</code>	Manual ECO Commands
<code>create_boundary_model</code> <code>-skip_disabled</code> <code>-include_userpins</code> <code>-latch_effort_low</code>	Timing Analysis Commands
<code>create_generated_clock</code> <code>-preinvert</code>	Timing Constraint Commands
<code>create_path_category</code> <code>-from_inst_group</code> <code>-to_inst_group</code>	Timing Debug Commands

<code>create_spice_deck</code> <code>-scale_mode</code>	Signal Integrity Commands
<code>delete_net</code> <code>-module</code>	Manual ECO Commands
<code>reset_timing_derate</code> <code>-transition</code>	Timing Analysis Commands
<code>report_constraint</code> <code>-drv_abs_margin</code> <code>-drv_ratio_margin</code>	Timing Analysis Commands
<code>report_case_analysis</code> <code>-include_function</code> <code>-include_ref_name</code>	Timing Analysis Commands
<code>report_noise</code> <code>-merge</code>	Signal Integrity Commands
<code>report_noise_propagation</code> <code>-style</code>	Signal Integrity Commands
<code>set_eco_opt_mode</code> <code>-high_effort_cell_list</code> <code>-smart_mmmc_force_keep_hold_view_list</code> <code>-</code> <code>smart_mmmc_force_keep_setup_view_list</code>	Signoff ECO Commands
<code>set_lib_binding_by_voltage</code> <code>-early_dynamic_power_net_voltages</code> <code>-late_dynamic_power_net_voltages</code>	Low Power Commands
<code>set_path_adjust_group</code> <code>-fall</code> <code>-fall_from</code> <code>-fall_through</code> <code>-fall_to</code> <code>-rise</code> <code>-rise_from</code> <code>-rise_through</code> <code>-rise_to</code>	Timing Analysis Commands

<code>set_sense</code> <code>-clock_leaf</code>	Timing Constraint Commands
<code>set_si_mode</code> <code>-virtual_inter_die_attacker_slew</code>	Signal Integrity Commands
<code>set_wire_variation_mode</code> <code>-wire_cell_variation_correlation</code> <code>-use_nsigma_coupling_capacitance</code>	Signal Integrity Commands
<code>write_ipd_scope</code> <code>-include_derates</code> <code>-include_constraints</code>	Timing Analysis Commands

Obsolete Commands

Removed from Software

The following command has been removed from the software:

- `reset_sdc`

Obsolete Global Variables

Removed from Software

The following global variable has been removed from the software.

- `timing_cpvr_enable_mismatch_transition_mode`

Obsolete Command Parameters

Removed from Software

The following obsolete text command parameters have been removed from the software.

- `create_boundary_model -validate`

Default Behavior Changes

- [Timing Analysis](#)
 - [Timing Report Format Default Change in PBA Mode](#)

Release 26.10 Enhancements

The following default changes have been made.

Timing Analysis

Timing Report Format Default Change in PBA Mode

Previous Behavior	New Behavior
Earlier, by default timing reports (<code>report_timing</code> command) did not display PBA column headers.	In this release, by default, the timing reports will display PBA column headers for nsigma, mean, sigma, and standard deviation details.
	Impact: This default change might impact parsing of reports in some cases.

Timing Modeling

- [Release 26.10 Enhancements](#)
 - [Added New Global Variables](#)

Release 26.10 Enhancements

The following enhancements have been made.

Added New Global Variables

The following global variables have been added:

- `timing_extract_model_half_cycle_non_interface_paths`: When set to `true`, the software models half-cycle non-interface paths as MPW (minimum pulse width) checks in ETM. By default, this global variable is set to `false`.
- `timing_extract_model_enable_output_pin_attributes`: Specifies the attributes (e.g., `always_on`, `is_isolated`) to enable on output pins during ETM generation. You can use this global variable when backward compatibility with older ETM behavior is required. By default, this global variable is set to `none`.
- `timing_extract_model_synchronous_clocks_as_skew_checks`: When set to `true`, the software additionally captures constraints in ETM for reg-to-reg paths, which are launched and captured by different synchronous clocks within the block. The software converts the slack of such paths into a skew-rising/falling constraint requirement between the two clock pins at the IP boundary. By default, this global variable is set to `false`.
- `timing_extract_model_write_function_for_combinational_arcs`: When set to `true`, the software writes the function attribute on ports with combinational arcs. By default, this global variable is set to `false`.

Delay Calculation

- Release 26.10 Enhancements
 - Support for Derate-Based Advanced Multi-Input Switching (AMIS-Derate) Analysis
 - Ability to Set Advanced Variation Settings
 - Ability to Scale Via Resistance using Scaling Factor
 - Added New `set_wire_variation_mode` Command Parameters

Release 26.10 Enhancements

The following enhancements have been made.

Support for Derate-Based Advanced Multi-Input Switching (AMIS-Derate) Analysis

A simulation-based advanced multiple-input switching (AMIS-Simulation) analysis can model MIS effects on both delay and slew. As a result of modeling MIS effects on slew, enabling AMIS-simulation can be very pessimistic. Due to the detailed simulation performed for each instance, the runtime overhead can be high.

To mitigate these shortcomings, the AMIS-derate feature has been introduced. AMIS-derate pre-characterizes MIS impact on multi-input cells as delay derates for different slew/load combinations and uses these derates in timing analysis. AMIS-derate applies MIS impact only to the base delay component. AMIS-derate maintains accuracy similar to AMIS-simulation and provides better runtime, debuggability, and reporting capabilities.

- To enable derate-based MIS, you can use the following setting:

- `set_analysis_mode -multi_input_switching_mode 3`

When MIS-derate is enabled, Tempus will automatically disable AMIS-simulation.

- To generate MIS-derate reports (in PBA mode only), the `report_timing -format` parameter supports the `retime_mis_derate` option.
- The `get_property` (and `report_property`) command supports the following new properties:

Property	Added to Object
<code>has_mis_fall</code>	◦ <code>timing_arc</code> ◦ <code>lib_timing_arc</code>
<code>has_mis_rise</code>	◦ <code>timing_arc</code> ◦ <code>lib_timing_arc</code>
<code>mis_alignments_factor_min_rise</code>	<code>lib_cell</code>
<code>mis_alignments_factor_min_fall</code>	<code>lib_cell</code>

For more information on AMIS-Derate, refer to "[Derate-Based Advanced Multi-Input Switching \(AMIS-Derate\) Analysis](#)" section in the *Tempus User Guide*.

Ability to Set Advanced Variation Settings

The Tempus software allows you to set up advanced variation settings. The following new command has been added:

- `set_advanced_variation_mode`: Sets the advanced variation scope to configure generic effect specific settings.

Ability to Scale Via Resistance using Scaling Factor

The Tempus software allows you to scale via resistance from SPEF, which is based on the user-specified scaling factor. The following new commands have been added:

- `set_via_variation_mode`
- `get_via_variation_mode`

The `set_via_variation_mode` command allows you to specify per via sigma multiplier and handles the advanced via variation scaling.

Added New `set_wire_variation_mode` Command Parameters

The following `set_wire_variation_mode` command parameters have been added:

- `-wire_cell_variation_correlation {true | false}`: Controls whether to RSS or add the random cell variation and wire variation for arrivals and slack. *Default:* `false`
- `-use_nsigma_coupling_capacitance {true | false}`: Allows to use the mean+nSigma coupling capacitance. *Default:* `false`

Timing Analysis and Reporting

- [Timing Enhancements](#)
- [Signal Integrity Enhancements](#)
- [Report Enhancements](#)

Timing Enhancements

- Release 26.10 Enhancements
 - Timing Analysis Enhancements
 - Support for Hierarchical Timing Context Analysis
 - Added New Global Variables
 - Added New timing_spatial_derate_distance_mode Option
 - Timing Constraint Enhancements
 - set_path_adjust_group Command Enhancements
 - Added New create_generated_clock Parameter
 - timing_generated_clocks_inherit_ideal_latency Global Variable Enhanced
 - Added New Parameters
 - Ability to Query Clock Pins and Clock Source Latency
 - IPD Analysis Enhancements
 - Enhanced write_ipd_scope Command
 - Scope Analysis Enhancements
 - create_boundary_model Command Enhancements

Release 26.10 Enhancements

The following enhancements have been made.

Timing Analysis Enhancements

Support for Hierarchical Timing Context Analysis

The Tempus hierarchical timing context feature provides top-down analysis techniques to model accurate block boundary constraints from the top hierarchy or top STA environment. Tempus provides a bottom-up (Boundary Models) and top-down (Context Models) hierarchical timing analysis/closure solution to address challenges encountered during timing and glitch analysis/closure of large hierarchical designs. The context models for lower-level modules are generated from top level analysis to perform accurate block level timing and glitch analysis. The context data is generated in a proprietary format and can be read with Tempus only.

To use this feature, the following setting is required:

```
set_analysis_mode -analysisType onChipVariation
```

The context analysis feature is not supported in best-case worst-case (BcWc) timing analysis mode.

The following features have been added:

New Commands

- `set_timing_context_clock_mapping`
- `set_timing_context_view_mapping`
- `report_timing_context_clock_mapping`
- `compare_timing_context`

New Global Variables

- `timing_hierarchical_context_continue_on_top_block_clock_mismatch`: Allows loading of the timing context model with unmapped clocks. *Default:* `false`
- `timing_enable_hier_context_unmapped_clock_analysis`: When set to `true`, for unmapped block clocks, the software enables timing analysis based on the block SDC constraints set on block ports. *Default:* `false`
- `timing_context_port_based_clock_mapping`: When set to `true`, clock mapping is done depending solely on clock property of the top and block clock for each clock and data port separately. *Default:* `false`
- `timing_context_apply_port_sdc_exceptions`: When set to `true`, the software will consider path

exceptions on block constraints in the hierarchical context flow. *Default:* `false`

For more information, refer to "[Hierarchical Timing Context Analysis](#)" section in the *Tempus User Guide*.

Added New Global Variables

The following new global variables have been added:

- `timing_consider_late_arrival_for_ssi_derate`: When set to `true`, the software uses the late arrival time for window overlapping evaluation during simultaneous switching input (SSI) analysis. *Default:* `false`
- `timing_enable_hierarchical_get_nets_support`: When set to `true`, the software enables the hierarchical segment support for the `get_nets` command. *Default:* `false`
- `timing_disable_get_clocks_for_current_instance`: When set to `true`, the software disables the existing `current_instance` command support on the `get_clocks` command. *Default:* `false`
- `timing_ipd_xdomain_path_filtering_threshold`: When this global variable is used, the specified threshold (in ns) will be used in the path filtering criteria during native inter-power domain (xdomain) solution. *Default:* `0.003`

Added New `timing_spatial_derate_distance_mode` Option

By default, spatial derate calculation uses the `chip_size` mode in GBA, which is conservative. The new arc bounding box mode has been introduced in Tempus STA, which reduces GBA spatial derate pessimism by limiting bounding box calculations to all timing paths through an instance. The reduced GBA-PBA gap and reduced GBA pessimism result in improved PBA runtime.

The following new option has been added to the `timing_spatial_derate_distance_mode` global variable:

- `arc_bounding_box`

You can use this option when

the `timing_enable_pessimistic_cpvr_for_reconvergent_clock_paths` global variable is set to `true`.

Timing Constraint Enhancements

`set_path_adjust_group` Command Enhancements

The `set_path_adjust_group` command supports the following edge-specific parameters:

- `-fall`: Applies `set_path_adjust_group` at the falling edge of the signal on the path endpoints.
- `-fall_from <from_list>`: Applies `set_path_adjust_group` at the falling edge of the signal on the

from pins.

- `-fall_through <through_list>`: Applies `set_path_adjust_group` at the falling edge of the signal on the through pins.
- `-fall_to <to_list>`: Applies `set_path_adjust_group` at the falling edge of the signal on the to pins.
- `-rise`: Applies `set_path_adjust_group` at the rising edge of the signal on the path endpoints.
- `-rise_from <from_list>`: Applies `set_path_adjust_group` at the rising edge of the signal on the from pins.
- `-rise_through <through_list>`: Applies `set_path_adjust_group` at the rising edge of the signal on the through pins.
- `-rise_to <to_list>`: Applies `set_path_adjust_group` at the rising edge of the signal on the to pins.

Added New `create_generated_clock` Parameter

The following `create_generated_clock` parameter has been added:

- `-preinvert`: Inverts the polarity of a master clock at the specified source pin.

`timing_generated_clocks_inherit_ideal_latency` Global Variable Enhanced

Earlier, when the `set_clock_latency` command was used to define network latency for ideal clock mode at intermediate pins of a clock network, Tempus ignored the latency. In this release, the `timing_generated_clocks_inherit_ideal_latency` global variable has been enhanced to accept Enum values that also allow including the closest applied network latency for ideal inheritance. The following new options have been added:

- `actual_partial_latency_path`: Uses the network latency of the master clock asserted closest to the target pin and propagated delays for the path from this point to the assertion point of the generated clock.
- `ideal_partial_latency_path`: Uses the network latency of the master clock asserted closest to the target pin and ideal delays for the path from this point to the assertion point of the generated clock.

Note: When the `-source` parameter is used to specify the latency of a generated clock, the software completely overrides any inherited latency.

By default, this global variable is set to `false`.

Added New Parameters

The following parameters have been added:

- `set_sense -clock_leaf`: When specified, the software propagates clock phases to the specified pin.
- `reset_timing_derate -transition`: Resets the transition derate values.

Ability to Query Clock Pins and Clock Source Latency

The `get_property` (and `report_property`) command has been enhanced to query on clock source objects and latency. The following new properties have been added:

Property	Object Type	Description
<code>source_latency_pins</code>	clock	Returns a collection of pins/ports in the source latency network of a generated clock. This property is undefined for master clocks. You can index by clock name using the <code>get_property -clock</code> parameter. <i>Return type</i> : collection
<code>is_clock_source_latency</code>	pin port net	Returns a value of <code>true</code> if a pin/port or any of the leaf pins of a net is part of a clock source latency network. <i>Return Type</i> : Boolean

<code>is_ref_clock_pin</code>	<code>pin</code> <code>lib_pin</code>	Returns a value of <code>true</code> for: - Valid instance pin object and on an active clock pin where a clock signal reaches, and sequential cell was not disabled by disabled timing arcs or case analysis. Any pin with a clock check or trigger arc will also be reported if an active clock phase is present on that pin. - Clock pin on an inferred clock-gating cell. - Valid <code>lib_pin</code> object, if there is at least one instance of this <code>lib_pin</code> in the design, and if the <code>is_clock_pin</code> property is set to <code>true</code>. - Returns a value of <code>false</code> if clock propagation is disabled with the <code>set_disable_timing</code> or <code>set_sense - stop_propagation</code> command. <i>Return Type:</i> Boolean
<code>is_clock_source_pin</code>	<code>pin</code> <code>port</code>	Returns a value of <code>true</code> for a valid instance pin or port object, if a clock is declared with that clock source at the pin/port. <i>Return Type:</i> Boolean
<code>is_clock_pin</code>	<code>lib_pin</code>	Returns a value of <code>true</code> if at least one of the timing arcs for setup/hold exists. Also, if equivalent timing constraints are present in the library definition of the library pin. <i>Return Type:</i> Boolean

IPD Analysis Enhancements

Enhanced `write_ipd_scope` Command

Tempus allows the identification of inter-power-domain (IPD) paths with more than two power-domain crossings. The `write_ipd_scope` command has been enhanced to support multiple power-domain paths during early IPD checks. The following new options have been added to the `-early_structural_checks` parameter:

- `multiple_pd_data_paths`
- `multiple_pd_clock_paths`
- `multiple_pd_paths`

The following `write_ipd_scope` command parameters have been added:

- `-include_constraints`: Allows to write IPD scope with victim constraints.
- `-include_derates`: Allows to write IPD scope with timing derates.

For more information, refer to the [Inter Power Domain \(IPD\) Analysis](#) chapter in the *Tempus User Guide*.

Scope Analysis Enhancements

`create_boundary_model` Command Enhancements

The following `create_boundary_model` command parameters have been added:

- `-skip_disabled`: Skips tracing of logic through disabled arcs (user or constant disabled) in the boundary model.
- `-include_userpins`: Includes the specified pins in the boundary model. No fanin/fanout tracing is performed for the specified pins.
- `-latch_effort_low`: Skips tracing of logic through latch aggressors that are not part of the interface logic. The aggressors in the fanout of latch aggressors will use an infinite timing window. By default, the entire fanin cone of latch aggressors is preserved to ensure accurate latency adjustment, which may increase the size of the boundary model.

The following parameters have been removed:

- `-absolute_tolerance`
- `-exclude_constraints`
- `-include_full_netlist`
- `-validate_localCpu`

Signal Integrity Enhancements

- [Release 26.10 Enhancements](#)
 - [Added merge_noise_reports Command](#)
 - [Added New Command Parameters](#)

Release 26.10 Enhancements

The following enhancements have been made.

Added merge_noise_reports Command

The Tempus software allows you to generate merged noise reports for the specified views. The following new command has been added:

- `merge_noise_reports`

Added New Command Parameters

The following new parameters have been added:

- `report_noise_propagation -style {default | extended}`: Provides an enhanced report format that shows new fields - UDN and User annotated. You can specify the following options:
 - `default`: Prints the ROP column and VDD (in mV) in the output report.
 - `extended`: Reports the drivers with annotated glitches as propagation sources.
- `set_si_mode -virtual_inter_die_attacker_slew <value>`: Specifies the virtual inter-die attacker slew value. *Default:* 2.5e-11
- `set_si_mode -enable_advanced_flow {true | false}`: Enables advanced SI analysis. *Default:* false
- `report_noise -merge`: Generates a merged report that shows only the worst glitch slack for each pin level combination across views.
- `create_spice_deck -scale_mode string`: Allows you to specify the UMC_URI scale mode option. *Default:* effective

Report Enhancements

- [Release 26.10 Enhancements](#)
 - [Added New Report Global Variables](#)
 - [report_constraint Command Enhancements](#)
 - [report_timing Collection Report Output Enhancements](#)
 - [Added New report_case_analysis Parameters](#)

Release 26.10 Enhancements

The following enhancements have been made.

Added New Report Global Variables

The following global variables have been added:

- `timing_property_is_clock_pin_consider_clock_source`: When set to `true`, the software considers the clock target pins as clock pins. By default, this global variable is set to `true`.
- `timing_report_constraint_enable_drv_on_ideal_network`: Enables DRV reporting of ideal nets. By default, this global variable is set to `true`.
- `timing_report_skip_icg_from_all_fanin_fanout_collection`: When set to `true`, the software skips ICG clock pins from the `all_fanin` (or `all_fanout`) command collection when queried with startpoints/endpoints. By default, this global variable is set to `false`.
- `timing_report_check_timing_enable_clock_crossing_verbose_clock_groups`: When set to `true`, the software displays clock groups (if present) in the `check_timing -check_only clock_crossing -verbose` command output. By default, this global variable is set to `false`.
- `timing_report_enable_qpin_as_start_for_begin_end_pair`: Enables reporting of multiple paths with different q-pins having the same clock pin. By default, this global variable is set to `false`.
- `timing_report_enable_write_twf_third_party_for_input_pins`: When set to `true`, the software writes timing information for input pins in third-party TWF file. By default, this global variable is set to `false`.
- `timing_report_print_normalized_slack`: When set to `true`, the software displays normalized slack based on the worst slack of different views. By default, this global variable is set to `false`.

- `timing_report_limit_max_path_count`: Sets the maximum number of violations (of data/clock checks) to be reported. By default, this global variable is set to `2000000` (integer).

report_constraint Command Enhancements

3214912: Make `-drv_abs_margin` and `-drv_ratio_margin` public in `report_constraint` command.

The following `report_constraint` command enhancements have been made:

- Added new parameters:
 - `-drv_abs_margin`: Specifies the absolute DRV margin value applied to obtain the threshold transition.
 - `-drv_ratio_margin`: Specifies the ratio DRV margin value applied to obtain the threshold transition.
- Added new `-drv_fields` option:

`hnet`: Reports hierarchical nets. When the `timing_enable_hierarchical_get_nets_support` global variable is set to `true`, the software enables support for displaying hierarchical net names.

report_timing Collection Report Output Enhancements

The timing report Retime Analysis header has been enhanced for collection reports. When the `report_timing -path_type exhaustive -collection` command is used, the following tags will be printed in the report output:

- `PBA COLL`: Shows a collection of paths for which retime analysis is performed while creating the collection.
- `GBA COLL`: Shows a collection of paths for which retime analysis is not performed.

When the `report_timing -collection` parameter is specified, the GBA and PBA slack values are not shown separately by default.

Added New report_case_analysis Parameters

The following `report_case_analysis` parameters have been added:

- `-include_function`: Displays the pin function of a pin. This must be specified with the `-propagated` parameter.
- `-include_ref_name`: Displays the reference name of an instance in the report.

Distributed Static Timing Analysis

- [Release 26.10 Enhancements](#)
 - [Supported Parameters in DSTA](#)

Release 26.10 Enhancements

The following enhancements have been made.

Supported Parameters in DSTA

The following parameters are now supported in DSTA:

- `set_license_check -checkout`
- `set_license_check -wait`

Manual ECO

- [Release 26.10 Enhancements](#)
 - [Added New add_net Parameter](#)
 - [Added New delete_net Parameter](#)

Release 26.10 Enhancements

The following enhancements have been made:

Added New `add_net` Parameter

The following new `add_net` parameter has been added:

- `-module_based`: Adds the new net to the specified Verilog module.

Added New `delete_net` Parameter

The following new `delete_net` parameter has been added:

- `-module_based`: Deletes the net from the specified Verilog module.

Signoff ECO

- [Release 26.10 Enhancements](#)
 - [set_eco_opt_mode Command Enhancements](#)

Release 26.10 Enhancements

The following enhancements have been made:

set_eco_opt_mode Command Enhancements

The following new `set_eco_opt_mode` parameters have been added:

- `-smart_mmmc_force_keep_hold_view_list <string>`: List of hold views, which have to be retained after SmartMMMC view compaction. These views represent the minimum hold views retained by the system, on top of which additional views can be retained.
- `-smart_mmmc_force_keep_setup_view_list <string>`: List of setup views, which have to be retained after SmartMMMC view compaction. These views represent the minimum setup views retained by the system, on top of which additional views can be retained.
- `-high_effort_cell_list <list of cells>`: Enables the controlled utilization of high-leakage or high-power cells during signoff optimization. These cells are typically classified as `dont_use` or `dont_touch` due to their power characteristics; however, they may be beneficial in specific scenarios. When this parameter is specified, ensure that the `don't_use` constraint is set to `false` for cells included in `-high_effort_cell_list`.
 - In setup timing optimization, the cells are considered only if they are essential to meet timing requirements without causing a significant increase in power consumption.
 - In power optimization, the cells may be inserted if their usage leads to an overall decrease in power usage, such as enhancements in fanin/fanout cone.

This parameter allows treating the specified cells as high-effort resources, using them only when necessary to improve timing or power efficiency, while maintaining the broader design objective of reducing power impact.

Other Enhancements

- Release 26.10 Enhancements
 - Voltage-Aware Dynamic Power Library Binding Enhancements
 - Enhanced Path Categorization with Instance Group Filtering
 - Enhancement to Restrict Saving the Database in the ECO Batch Mode
 - Added New Global Variable
 - Added New Global Variable
 - Clock Tree Debugger Enhancements
 - Support for Coloring Based on Maximum Fanout Constraint
 - Support for Displaying Partition Hierarchy
 - Signal Edge Coloring Enhanced
 - Support for Displaying Skew Group Source Pin
 - Support for Zooming into a Selected Item in the Clock Tree Viewer
 - Support for Displaying Skew Group Source Pin
 - Quick Access to Object Attributes
 - Color Map Changes for Setup and Hold Slack

Release 26.10 Enhancements

The following enhancements have been made:

Voltage-Aware Dynamic Power Library Binding Enhancements

In this release, the `set_lib_binding_by_voltage` command has been enhanced with the addition of the following new parameters to improve voltage awareness during dynamic power analysis at different stages of the flow:

- `-early_dynamic_power_net_voltages`
- `-late_dynamic_power_net_voltages`

These parameters enable the software to consider power-net voltage information during early-stage and late-stage dynamic power analysis, respectively. By enabling voltage-aware library binding at appropriate phases of the flow, this enhancement provides flexibility and control over how voltage information is applied during dynamic power analysis across the design flow.

Enhanced Path Categorization with Instance Group Filtering

The `create_path_category` command has been enhanced to improve timing path selection with the introduction of the following two new parameters:

- `-from_inst_group`: Specifies the source instance group from which timing paths need to begin. When this parameter is specified, the software considers only those timing paths whose start points belong to the specified instance group.
- `-to_inst_group`: Specifies the destination instance group where timing paths need to end. When this parameter is specified, the software includes only those timing paths whose end points belong to the specified instance group.

Enhancement to Restrict Saving the Database in the ECO Batch Mode

A new global, `save_db_restrict_under_eco_batch_mode`, has been introduced. You can use this global to restrict saving the database in the ECO batch mode.

Added New Global Variable

To override updated macros in the database when LEF files are specified during the restore flow, a new global variable, `restore_db_keep_updated_macro_lefs`, has been added. When set to `true`, the LEF files of updated macros that are saved in the database will be used, even if new LEF files are specified for these macros with the `read_design -lef_files` command. When set to `false`, the LEF files of updated macros in the database will be ignored. Only the LEF files that are specified with the `read_design -lef_files` command will be updated.

Added New Global Variable

The following new global variable has been added:

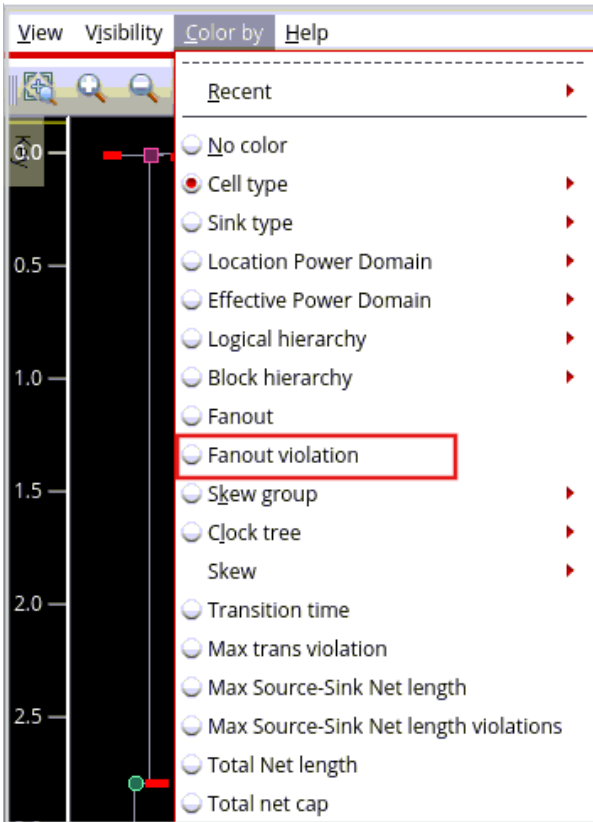
- `timing_library_ignore_combinational_arc_mismatch_tables`: When set to `true`, the software will give preference to `timing_type` over arc tables. This will change the inferred arcs based on the `timing_type` (liberty format) or timing tables settings. A warning message will be displayed when the preference (`timing_type` or arc tables) is changed.

By default, this global variable is set to `false`.

Clock Tree Debugger Enhancements

Support for Coloring Based on Maximum Fanout Constraint

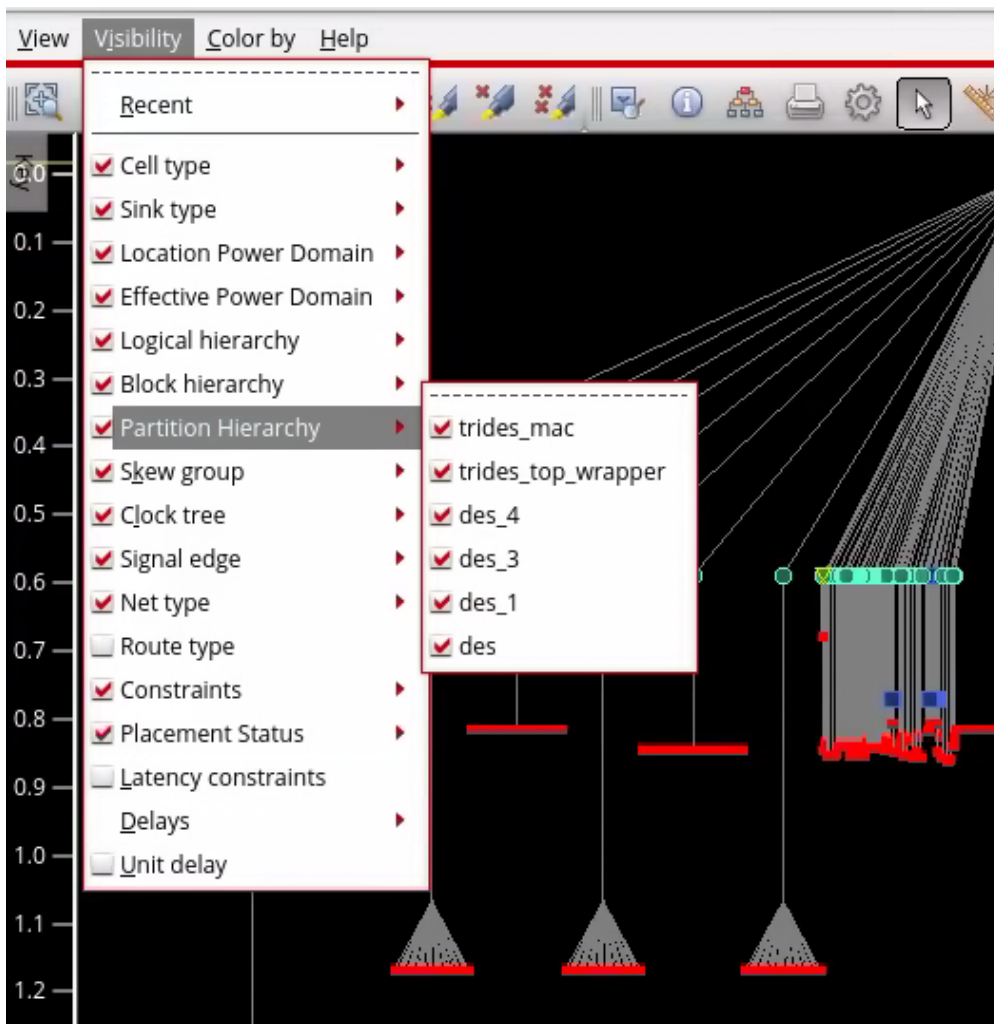
The CTD is enhanced to support coloring selected nodes based on the maximum fanout constraint. For this feature, a new option, *Fanout violation*, is added to the *Color by* menu and the *Control* panel. The tooltip shows the violation amount and target value.



For details, see the [Clock Menu](#) chapter in *Tempus Menu Reference*.

Support for Displaying Partition Hierarchy

A new option called *Partition Hierarchy* has been provided in the *Visibility* menu, *Color by* menu, and *Control* panel. This option provides a color category for each partition in the design. Each marker is drawn with the color associated with that partition.



This option is only shown in these menus for designs with partitions configured and is available in unit delay and timing-based modes.

Signal Edge Coloring Enhanced

The *Signal edge* option in the *Visibility* menu, the *Color by* menu, and the Control panel provide options to view and color signal edges for cells and nets. In this release, the signal edge feature is enhanced in the following ways:

- The cell coloring was previously only based on the edges found on the input and output nets. New cell coloring is based on the more accurate timing arcs.
- The *Cell* and *Net* items of the *Signal edge* option now have no color associated with them in the tree view. Only the options available within each subgroup have color options.
- The default coloring is changed to make them more colorblind friendly.
- The *Root* option is removed from the *Cell* subgroup. This option was used for both non-generated

and generated clock roots. It is replaced by a new *Unknown* option that includes non-generated root cells. The generated root cells are now colored according to the *Simple/Inverting/Non-unate* options.

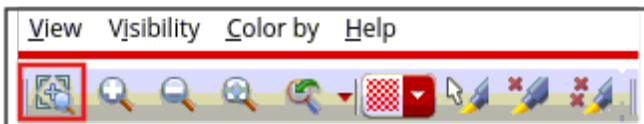
- The *Unknown* option is also added in the *Net* subgroup. It is used for nets with no skew groups.
- The *Net* subgroup coloring is enhanced to include nets connected to generator path nodes.

Support for Displaying Skew Group Source Pin

When displaying a single skew group, the source pin is typically not shown because CTD markers represent input pins, while skew groups usually start at output pins. Starting from this release, CTD is enhanced so that when coloring or displaying by skew group, a marker is treated as part of the skew group if either its input pin or the output pin of the corresponding clock node belongs to that skew group. This ensures that it remains visible when other skew groups are hidden.

Support for Zooming into a Selected Item in the Clock Tree Viewer

In this release, a new toolbar widget, *Zoom Selected*, is added to the row of widgets below the CTD menu bar. This widget allows you to zoom into a selected item in the Clock Tree Viewer without opening the *Set Preference* form. The equivalent bindkey is `Ctrl+Z`.



As part of this enhancement, the *Enable Zoom selected* option in the *Set Preference* form is renamed to *Enable Zoom layout selected*. The equivalent bindkey provided is `Ctrl+E`.

Support for Displaying Skew Group Source Pin

When displaying a single skew group, the source pin is typically not shown because CTD markers represent input pins, while skew groups usually start at output pins. Starting from this release, CTD is enhanced so that when coloring or displaying by skew group, a marker is treated as part of the skew group if either its input pin or the output pin of the corresponding clock node belongs to that skew group. This ensures that it remains visible when other skew groups are hidden.

Quick Access to Object Attributes

You can now view object attributes directly from the Clock Tree Viewer using the `⌵` bindkey. Select any object and press `⌵` to open the *Object Attributes* form, providing faster access to detailed object information without navigating menus.

Color Map Changes for Setup and Hold Slack

The color map represents the slack range from negative to positive values. In this release, it is enhanced so that white indicates the most negative slack, transitioning to red at zero slack. Positive slack is shown starting in green and progressing to purple, which represents the highest positive slack.

